DocQuorum

Agentic document extraction with open models:
the pipeline on real benchmark documents

October 2026

Every example is a real benchmark document and a stored pipeline output; no model was called to make these slides.

Outline

1. Task and pipeline
2. Walkthrough on real documents
   inputs, routing, parsing, readers, deterministic checks, selection by quorum, output
3. Results on six benchmarks: places, cost, speed; zoom re-reading
4. Summary
5. Complete results, one table per benchmark

Setting

Open-weight models only: Qwen3.8-27B (FP8) for reading and MinerU2.5 for parsing scans, served on a single GPU. Each benchmark is scored with its own official scorer.

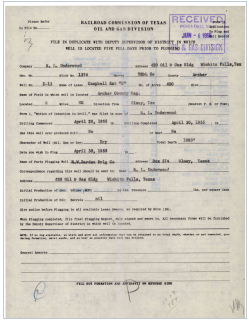
Problem statement

Task

Input: a document D (PDF or image), a JSON Schema S or a JSON template, and optionally a task description T.

Output: one JSON object Y that conforms to S, every value supported by D, absent values `null`.

Optional task (real example, receipt schema):
"A single point-of-sale purchase RECEIPT (retail or restaurant). DOC-TYPE / SCOPE RULE: if one file bundles MULTIPLE . . . "


RAILROAD COMMISSION OF TEXAS
OIL AND GAS DIVISION

RECEIVED

FILE IN DUPLICATE WITH DISTRICT SUPERVISOR OF DISTRICT IN WHICH WELL IS LOCATED FIVE FULL DAYS PRIOR TO PLUGGING

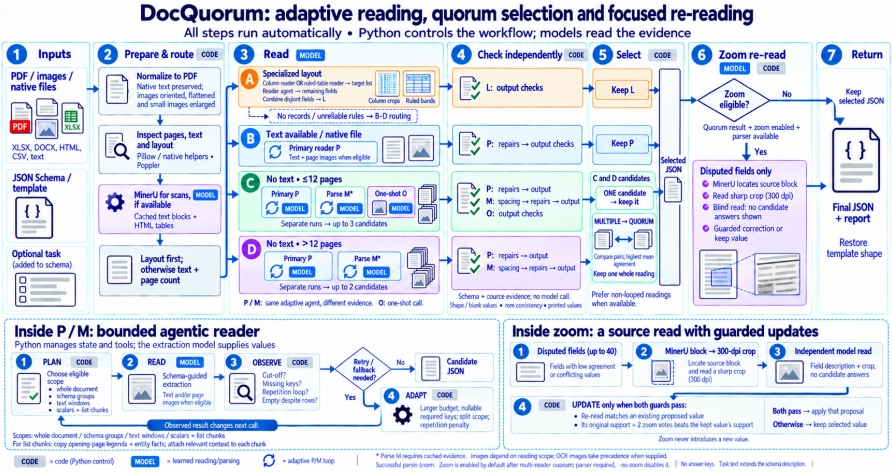
Scanned 1955 well-plugging application (Railroad Commission of Texas), 2 pages, no text layer.

JSON Schema (77 fields), excerpt

```
"p1_company":  {
  "anyOf":  [{"type":  "string"},
             {"type":  "null"}],
  "description":  "The 'Company' field
    in the identification block on
    page 1 ...  Extract verbatim as
    printed; null if blank.  Do NOT
    confuse with ..."
},
"p1_county":  {...},
"p1_total_depth":  {...},
...  74 further fields
```

ExtractBench short: 2A-9-160298_109929 (schema shortened); task text from the schema of hingham_lowes_receipt.

Pipeline overview



Pipeline figure (generated with Codex), describing the single-document entry point `docquorum.extract`; the lower panels show the bounded reader loop and guarded zoom updates (v1.3; v1.4 adds no step).

Step 2: the available evidence decides the route



A Layout reader

Unclaimed-property register
59 pages, 15,280 char./page



B Reader agent: text + images

County purchase order
1 page, 2,253 char./page



B Reader agent: text + images

W-14 well form
1 page, 252 char./page (OCR layer)



C Three readings, then quorum

1955 plugging application
2 pages, no embedded text



D Reader agent on the parse

Tax return
25 pages, no embedded text

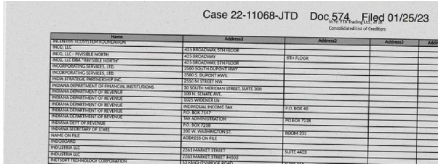
Routing rule

Layout first: pages printed as equal-width columns of records, or a ruled table on a scan, go to a layout reader A. Otherwise a PDF with embedded text (at least 50 characters per page; a digital PDF or an OCR layer) is read once B. A scan without text is parsed by MinerU and read three ways if it has at most 12 pages C, or read through its parse D.

ExtractBench documents; embedded characters per page measured with pdftotext.

Step 2: MinerU returns a scanned table as cells

Scanned page (detail)



MinerU output (raw HTML, abbreviated)

```
<tr><td>Name</td><td>Address1</td><td>Address2...<td>INCENTIVE
ECOSYSTEM FOUNDATION</td><td...<td>INCO, LLC</td><td>425
BROADWAY, 5TH FL...
```

The same rows as table cells (65 rows on this page)

Name	Address1	City	State	PostalCode
INCENTIVE ECOSYSTEM FOUNDATION				
INCO, LLC	425 BROADWAY, 5TH FLOOR			
INO, LLC - INAVAILABLE NORTH	425 BROADWAY	NEW YORK	NY	10013
INO, LLC DBA "INAVAILABLE NORTH"	425 BROADWAY, 5TH FLOOR	NEW YORK	NY	10013-2599
INCORPORATING SERVICES, LTD.	3500 SOUTH COOPONT HWY	NEW YORK	NY	10013

Columns stay aligned: an empty cell stays empty rather than pulling the next value into its place. Recognition errors remain, so the reader receives the page image together with the parse.

ExtractBench long: real_ftx_full_corrupted, page 47; cached MinerU2.5 parse.

Step 3, route A: the column reader



COLLINS RICHARD
EATON BETHINA
RT 1 BOX 220
JOHNSON JOHNEY B
C/O MAE JEAN JOHNSON,
RT 1 BOX 230
SHERRILL IVANNA
RT 1 BOX 1142

PROCTOR

BLEVINS JAMES DEE
RR2 BOX 556
SANDRA COTTONGAME
67000 S 367 RD

STILWELL

ADAIR TERI
RT 2 BOX 1155
BEASLEY ALAN C
470241 E 880 RD
BRD ALFRED JOSEPH
RT 2 BOX 271A
BREWER VENA MAUD
301 S CHEROKEE ST
BRIDGES INEZ
77150 HIGHWAY 59
BROWN CLIFFORD LEE
RT 4 BOX 2570
BROWN LISA
RT 3 BOX 2119
CRAWFORD BRADLEY
212 6TH PL
CRITTENDEN JESSIE
RT 6 BOX 1760
EUBANKS JR SHELBY
RT 4 BOX 1474
FOURKILLER HENRY
471958 E 810 RD

Records returned for column 1 (page 2)

name	locale
COLLINS RICHARD	null
EATON BETHINA	RT 1 BOX 220
JOHNSON JOHNEY B	C/O MAE JEAN JOHNSON, RT 1 BOX 230
SHERRILL IVANNA	RT 1 BOX 1142

7
columns detected on the page

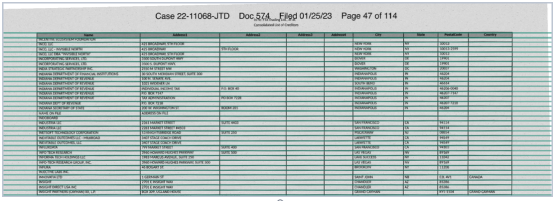
59
pages

26,675
records in the final output

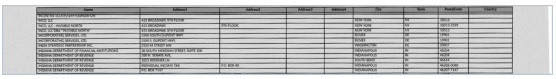
The trigger is page geometry only: at least three text columns of near-equal width. Each column is cut out and read top to bottom in one call; a record continued from the previous column is joined to it.

ExtractBench long: real_oklahoma_unclaimed_2024, page 2. Column boxes drawn with the pipeline's own geometry code; records from the stored final output.

Step 3, route A: the ruled-table reader



Printed rules detected after deskewing by 1.15° (green).



One call: the header row stacked on a band of 12 rows; the reply must contain exactly 12 entries.

ExtractBench long: real_ftx_full_corrupted, page 47. Rows from the stored final output.

67

rows counted from the rules

67

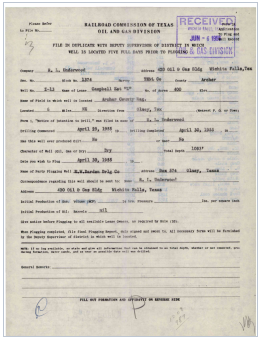
rows returned by the reader

Bands on page 47: $12 + 12 + 12 + 12 + 12 + 7$.

name	address_1	city	state
INCENTIVE ECOSYSTEM FOUNDATION	null	null	null
INCO, LLC	425 BROADWAY, 5TH FLOOR	NEW YORK	NY
INCO, LLC - INVISIBLE NORTH	425 BROADWAY	NEW YORK	NY

Fixing the number of entries per band prevents repeated identical rows from multiplying or disappearing.

Step 3, route C: three independent readings of one scan



field	P image reading	M parse reading	O one-shot
p1_company	R. L. Underwood	R. L. Underwood	null
p1_county	Archer	Archer	null
p1_well_no	Z-13	Z-13	null
p1_name_of_lease	Campbell Est "Z"	Campbell Est "Z"	null
p1_drilling_commenced	April 29, 1955	April 29, 1955	null
p1_party_plugging_well	R.W. Darden Dr'lg Co	R.W.Darden Drlg Co	null
p2_correspondence_name	R.L. Underwood	R.L.Underwood	null

Red: empty, or the two readings disagree (P and M differ only in spacing and punctuation).

P 59 / 77 fields filled

M 59 / 77 fields filled

O 1 / 77 fields filled

P: the reader agent on the page images. M: the reader agent on the MinerU parse and the images. O: the benchmark's own one-shot reader.

ExtractBench short: 2A-9-160298_109929; stored readings after their checks.

Step 4: checks that repair the structure of a reading

Check	Document	Observed change	Value F1
Rows copied into another list	Pueblo check register (medium)	**562 → 51 batch rows** Check 01033982 ($700.88) had also been filed as a batch with the same number and amount.	84.73 → 99.91
Identical records capped at the printed count	FTX creditor list, text layer (long)	**146 → 37 rows on page 59** Page 59 prints the line "NAME ON FILE" exactly 37 times.	84.47 → 90.77
Stacked lines folded into one record	Clinical adverse-event listing (long)	**1,499 → 668 records** Subject R1002: three printed lines became one event with its resolved date.	38.80 → 68.77
Template rows dropped	Texas W-14 well form (short)	**5 → 2 casing rows** Three rows carried only the printed row label (e.g. "Intermediate").	82.30 → 93.23

Examples are the cases each check was designed for. On a few short documents some repairs lowered the score slightly (e.g. duplicate removal on three election reports, −3.5 points in total).

Stored readings before and after each check, re-run with `docquorum/checks` and scored with ExtractBench's scorer. Each check uses only the schema and the document's own text; no model is called.

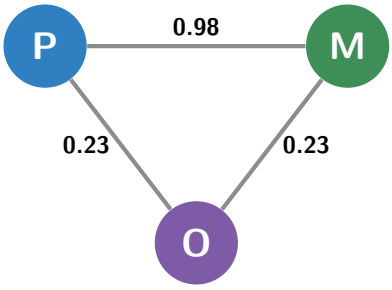
Step 4: checks that make values exactly what the page prints

Check	Document	Observed change	Value F1
Every schema field present	Caterpillar 330 spec sheet (short)	**missing keys → `null`** The reply omitted 7 optional fields in every specification row; the scorer counts an absent key as a miss.	61.48 → 95.74
A blank tick box is false	Texas P-4 form (short)	**6 boxes: `null` → `false`** The field description states "false if left blank".	91.67 → 96.67
Quotes as printed	Census statistical table (short)	**Bachelor's → Bachelor's** The page prints a typographic apostrophe; only the quote character is copied from the page.	60.12 → 62.58
Numbers as printed	Schedule K-1 (short, one-shot reading)	**33.333333 → 33.3333333** A rounded value is replaced by the single printed figure it restates (6 share fields).	85.42 → 97.92

Every replacement is a value that the document itself prints, or `null` where the schema admits null.

Stored readings before and after each check, re-run with `docquorum/checks` and scored with ExtractBench's scorer. Each check uses only the schema and the document's own text; no model is called.

Step 5: selection by quorum



Edge: agreement of two readings, measured with the benchmark's value scorer, recall-weighted (F_β, $\beta = 2$).

Reading	Mean agreement	
P image reading	0.604	**kept**
M parse reading	0.604	
O one-shot	0.218	

P and M tie; ties go to the reading listed first. The whole of P is returned; fields are never mixed across readings.

Agreement is not accuracy

Answer-key value F1 for this document (illustration only; never used by the pipeline): P 95.24, M 95.24, O 22.50. On the scanned county purchase order the quorum kept the one-shot reading (78.26), although the image reading scored higher (82.61).

ExtractBench short: 2A-9-160298_109929; agreement computed by `docquorum.quorum.pick` on the checked readings.

Step 6: one checked JSON, in the form requested

JSON Schema in, JSON out (1955 form, 77 fields)

```
{
  "p1_company":  "R. L. Underwood",
  "p1_address":  "420 Oil & Gas Bldg Wichita
Falls, Tex",
  "p1_county":  "Archer",
  "p1_well_no":  "Z-13",
  "p1_name_of_lease":  "Campbell Est \"Z\"",
  "p1_drilling_commenced":  "April 29, 1955",
  "p1_total_depth":  "1093'",
  ...  70 further fields
}
```

JSON template in, the same template filled



Template

```
{"menu":  [{"price":  "", "nm":  ""}],
 "total":  {"cashprice":  "",
            "total_price":  ""}}
```

Output (equal to the answer key)

```
{"menu":  [
   {"price":  "17500", "nm":  "J.STB PROMO"},
   {"price":  "46000", "nm":  "Y.B.BAT"},
   {"price":  "27500", "nm":  "Y.BASO PROM"},
 ],
 "total":  {"cashprice":  "91000",
            "total_price":  "91000"}}
```

Left: stored final output for 2A-9-160298_109929. Right: UNIKIE-BENCH Catering-Services test_1.jpg, quorum over six stored readings.

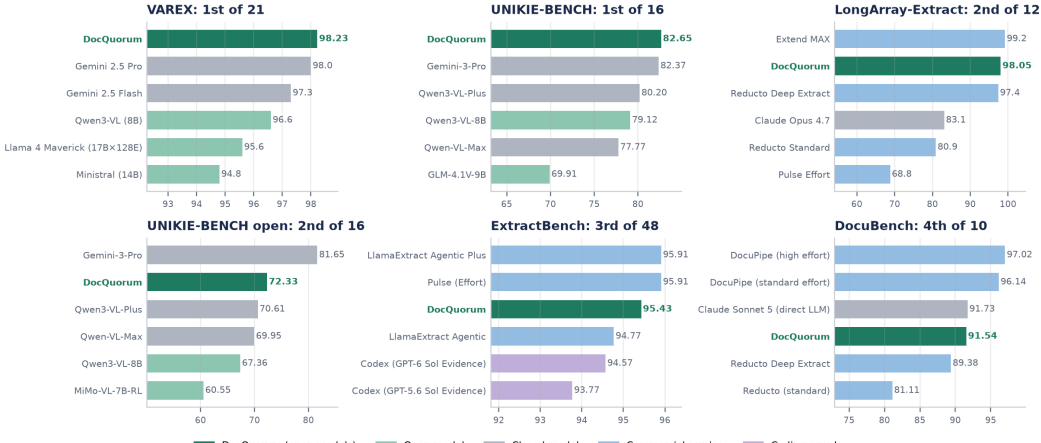
Results on six benchmarks

Benchmark	Score	Place	Ahead of	Calls	$ per document	s
VAREX 1,777 government forms	**98.23**	1st of 21	Gemini 2.5 Pro 98.0	3.7	0.021	25
UNIKIE-BENCH 4,472 images, 11 scenarios	**82.65**	1st of 16	Gemini-3-Pro 82.37	6.3	0.010	51
LongArray-Extract 45 long documents, 29,328 rows	**98.05**	2nd of 12	Reducto Deep Extract 97.4	100.3	0.676	1,466
UNIKIE-BENCH open 1,661 documents, 8 scenarios	**72.33**	2nd of 16	Qwen3-VL-Plus 70.61	4.6	0.018	121
ExtractBench 370 PDFs, 1 to 192 pages	**95.43**	3rd of 48	LlamaExtract Agentic 94.77	41.7	0.407	568
DocuBench 72 documents, 10 file types	**91.54**	4th of 10	Reducto Deep Extract 89.38	11.1	0.212	261

Scores: each benchmark's own scorer on the full test set; places count every published system. Open-weight models only (Qwen3.8-27B, MinerU2.5) on one GPU. Calls, $ and s: means per document over a random sample; $ prices the measured tokens at the Qwen3.8-27B list price.

Complete tables for every benchmark: the last section, and docs/results/TABLES.md. No result was submitted to a leaderboard.

Among the best published systems



Each benchmark's six best systems, DocQuorum among them, by the benchmark's own score; every row of each benchmark is in the complete tables (last section).

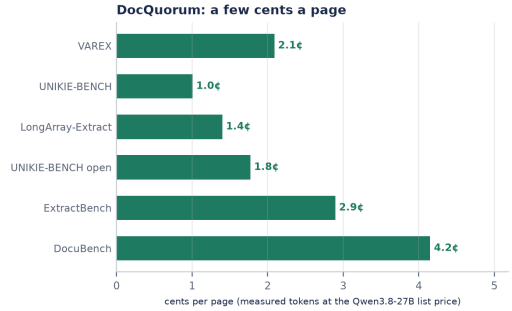
Ahead of open and closed models

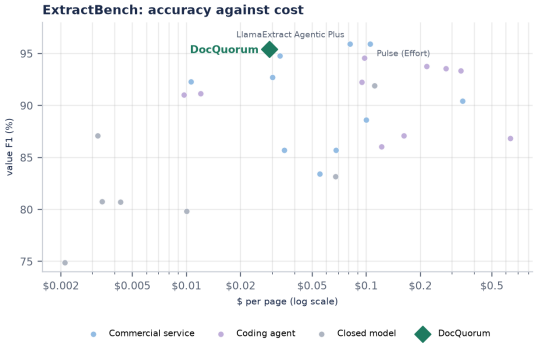


The gain comes from the pipeline, not a bigger model: the same Qwen3.8-27B called once scores **89.75** on ExtractBench (its leaderboard); through DocQuorum it scores **95.43**.

Grey or light green: the best published closed or open model on each benchmark; green: DocQuorum. LongArray-Extract and DocuBench publish no open models. Gains at each benchmark's published precision.

Low cost per page





On ExtractBench, the two systems above DocQuorum cost **$0.081** and **$0.105** a page; DocQuorum's tokens come to **$0.029** a page, and no published system is both cheaper and more accurate.

Left: means over a random sample of each benchmark (results/costs/). Right: every ExtractBench system the leaderboard prices (cost per page weighted by split, as the leaderboard computes it); DocQuorum's measured tokens at the Qwen3.8-27B list price ($0.50 / $3.00 per million input / output tokens). DocQuorum itself runs on one GPU, with no price per call.

Why a document costs little

One reading when the file has text

13 of 20 sampled DocuBench documents took a single model call, $0.009 each on average.

More readings only for scans

Scans get three readings, run at the same time; zoom re-reads only the disputed fields a new vote could change. A VAREX form takes 3.7 calls.

An open 27B model

$3 per million output tokens at list price: a tenth of GPT-5.5 ($30), a fifth of Claude Sonnet 5 ($15). On its own GPU, no price per call at all.

Accuracy from the pipeline

The same model called once scores 89.75 on ExtractBench; DocQuorum with it, 95.43: the extra calls buy the accuracy.

results/costs/: every measured document (calls, tokens, seconds); prices in docs/results/pricing.json. Long scans cost more than short files: they are read in chunks, once per reading.

v1.4: long scans read faster

The same documents, measured alone

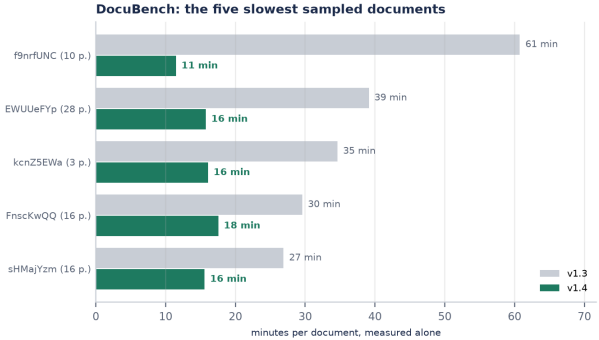
DocuBench (20): **2.4× faster**
12,564 s → 5,225 s

VAREX (30 forms): **1.6× faster**
1,178 s → 750 s

Loops stopped by the model server once one stretch of tokens repeats 40 times, not at the output limit.

A scan's readings at the same time, entering the quorum in the same order.

DocuBench, all 72 documents read again: **91.54**.



results/costs/v1.3 and results/costs: every measured document, before and after.

Zoom re-reading: a new step after the quorum

After the quorum keeps one reading, each field on which the readings disagree is found on the page and read once more from a sharp crop of that region.

This new reading sees none of the earlier answers. It changes the kept value only if it matches a value some reading already proposed, and it counts as two votes: it can overturn a lead of one reading, not a clear majority.

The vote rule was added after the third test, where five agreeing readings were overturned too often.

Three tests

VAREX: 98.10 → **98.30**
60 forms better, 25 worse

UNIKIE open: 71.07 → **72.33**
291 documents better, 138 worse

UNIKIE (six readings): 82.62 → 81.62;
82.65 with the vote rule

results/README.md: method, statistics, and every changed value checked against the answer key after the runs.

Summary

Results

1st on VAREX and UNIKIE-BENCH; 2nd on LongArray-Extract and UNIKIE open; 3rd of 48 on ExtractBench.

Ahead of every published open model, with open models on one GPU, at a few cents a page.

Zoom re-reading improved VAREX and UNIKIE open.

Method

Routing by the available evidence; an agentic reader that plans, reads and re-reads; layout readers for column flow and ruled tables; deterministic checks against the schema and the page; selection by quorum.

Limitations

ExtractBench served as the development set. Agreement among readings does not guarantee accuracy. Checks are triggered by schema wording and page text; a few repairs lower the score on some documents.

VAREX: complete results

#	System	Type	Score	ANLS	Flat	Nested	Table	Perfect
1	**DocQuorum (ours)**	Open models	**98.23**	**98.8**	**98.4**	**98.5**	**97.4**	**86.2**
2	Gemini 2.5 Pro	Closed model	98.0	98.5	97.3	98.2	97.7	82.8
3	Gemini 2.5 Flash	Closed model	97.3	97.9	96.9	97.7	96.5	81.0
4	Qwen3-VL (8B)	Open model	96.6	97.6	96.6	97.3	95.0	78.2
5	Llama 4 Maverick (17B×128E)	Open model	95.6	97.1	95.1	96.1	94.4	68.7
6	Ministral (14B)	Open model	94.8	96.5	94.8	95.1	94.0	66.0
6	GPT-4o	Closed model	94.8	97.1	95.1	95.3	93.5	64.3
8	Llama 4 Scout (17B×16E)	Open model	94.3	95.8	95.2	94.3	94.1	63.4
9	granite-vision-4.1-4b (4B)	Open model	94.2	96.4	95.2	94.9	92.3	65.2
10	NuExtract 2.0 (2B)	Open model	90.8	92.1	93.1	91.9	87.3	52.7
11	InternVL3.5 (2B)	Open model	85.6	89.1	89.6	86.1	82.7	42.4
12	granite-4.0-3b-vision (3B)	Open model	85.5	89.1	91.5	85.2	84.0	41.1
13	Qwen 2.5-VL (3B)	Open model	82.5	85.4	79.9	83.1	81.9	43.7
14	Gemma 3n (E2B)	Open model	71.0	78.1	76.7	71.0	68.8	19.0
15	MiniCPM-V4 (4B)	Open model	67.9	69.8	84.8	66.2	65.5	23.6
16	Gemma 3 (4B)	Open model	65.3	75.0	71.7	65.0	63.6	13.7
17	h2oVL Miss. (2B)	Open model	61.3	64.9	67.2	61.3	58.9	11.6
18	Qwen3-VL (2B)	Open model	34.2	34.7	93.3	26.8	29.0	29.3
18	h2oVL Miss. (800M)	Open model	34.2	36.1	35.8	33.2	36.0	5.9
20	InternVL3.5 (1B)	Open model	28.2	29.8	79.0	22.4	22.2	11.2
21	Qwen2-VL (2B)	Open model	9.7	10.0	16.4	9.0	8.9	4.3

Score: exact match (normalized) over 21,084 fields. ANLS: normalized ANLS; Flat, Nested, Table: exact match by form type (299, 1,146, 332 forms); Perfect: forms with every field right. Underlined: best in the column. Equal scores share a rank.

VAREX-Bench image-modality leaderboard (20 models, zero-shot; one decimal, so ranks compare at one decimal). DocQuorum: the final pipeline.

UNIKIE-BENCH: complete results

#	System	Type	Score	Ret.	Cat.	Com.	Acc.	Post.	Adm.	Edu.	Adv.	Tax	Med.	Nut.
1	**DocQuorum (ours)**	**Open models**	**82.65**	**90.63**	**88.13**	**53.51**	**79.29**	**85.65**	**87.00**	**85.67**	**86.59**	**82.16**	**78.33**	**92.22**
2	Gemini-3-Pro	Closed model	82.37	89.27	83.54	53.87	81.43	90.74	84.32	80.71	90.54	81.54	77.60	92.47
3	Qwen3-VL-Plus	Closed model	80.20	89.12	86.45	50.57	77.86	83.55	78.91	87.44	83.14	78.93	76.46	89.72
4	Qwen3-VL-8B	Open model	79.12	88.26	86.23	51.14	71.07	83.80	78.90	87.39	81.23	78.74	77.19	86.36
5	Qwen-VL-Max	Closed model	77.77	86.67	84.68	49.33	76.07	84.85	79.96	82.46	82.76	77.94	75.62	75.14
6	GLM-4.1V-9B	Open model	69.91	62.32	77.33	46.67	76.43	76.46	78.75	65.83	81.99	70.68	72.08	60.47
7	GPT-4o	Closed model	69.15	75.79	79.33	49.67	67.86	82.95	64.26	47.77	73.56	75.33	67.57	76.51
8	MiMo-VL-7B-RL	Open model	68.88	77.45	77.66	49.01	67.50	76.70	76.81	76.65	66.28	74.86	68.70	46.10
9	InternVL3.5-8B	Open model	67.70	77.23	72.23	41.64	67.14	73.90	72.10	63.08	73.08	72.81	65.94	65.53
10	GPT-5	Closed model	67.59	71.90	71.33	45.94	69.29	80.95	62.79	50.26	75.48	72.05	67.71	75.83
11	MiniCPM-V4.5-8B	Open model	67.15	68.37	74.91	45.91	67.76	83.60	71.89	63.21	79.31	69.57	70.00	44.18
12	Claude-Sonnet-4.5	Closed model	66.39	69.24	73.65	45.92	69.29	79.45	69.20	39.46	85.06	67.79	63.96	67.24
13	Kimi-VL-A3B	Open model	66.14	86.49	63.12	42.87	61.79	83.40	68.72	80.37	72.41	77.80	64.93	25.72
14	Gemma-3-12B	Open model	48.09	54.32	68.73	30.68	53.93	72.30	37.14	17.56	36.78	56.81	48.54	52.20
15	Ministral-3-8B	Open model	37.62	6.47	57.01	26.24	43.39	75.80	32.37	29.54	19.53	65.85	51.46	6.17
16	SmolVLM2-2.2B	Open model	23.69	17.80	57.22	28.52	18.92	34.98	8.63	3.87	27.16	34.97	18.00	10.51

Score: macro F1 over 11 scenarios. Ret. retail; Cat. catering services; Com. commercial; Acc. accommodation; Post. postal label; Adm. administrative; Edu. education; Adv. advertisement; Tax: tax-compliant; Med. medical services; Nut. nutrition label. Underlined: best in the column. Equal scores share a rank.

UNIKIE-BENCH (ACL 2026), Table 4 (15 published systems); every input capped at 1,605,632 pixels. DocQuorum: quorum over six readings from three open models, then zoom re-reading. DocQuorum: the final pipeline.

LongArray-Extract: complete results

#	System	Type	Score	Completed
1	Extend MAX	Commercial service	99.2	45/45
2	**DocQuorum (ours)**	**Open models**	**98.05**	**45/45**
3	Reducto Deep Extract	Commercial service	97.4	45/45
4	Claude Opus 4.7	Closed model	83.1	45/45
5	Reducto Standard	Commercial service	80.9	45/45
6	Pulse Effort	Commercial service	68.8	45/45
7	Pulse Auto	Commercial service	64.5	45/45
8	Gemini 3.1 Pro	Closed model	47.3	45/45
9	LlamaParse Standard	Commercial service	47.2	26/45
10	LlamaParse Agentic	Commercial service	34.7	35/45
11	GPT-5.5	Closed model	31.6	45/45
12	Gemini 3.5 Flash	Closed model	31.1	45/45

Score: macro document accuracy by the publisher's reference scorer (bank statements, clinical listings, indictments); a failed document counts as zero. Completed: documents answered. Equal scores share a rank.

Extend AI's LongArray-Extract results page (11 systems), which reports one decimal; ranks compare at one decimal. DocQuorum: the final pipeline.

UNIKIE-BENCH open: complete results

#	System	Type	Score	Ret. zh	Form zh	Inv. zh	Con. zh	Ret. en	Form en	Inv. en	Con. en
1	Gemini-3-Pro	Closed model	81.65	78.50	79.17	78.63	75.96	93.57	80.41	87.06	79.90
2	**DocQuorum (ours)**	**Open models**	**72.33**	**67.67**	**69.80**	**69.72**	**74.25**	**83.10**	**67.65**	**68.22**	**78.22**
3	Qwen3-VL-Plus	Closed model	70.61	71.14	69.87	69.59	70.95	68.27	66.82	73.45	74.81
4	Qwen-VL-Max	Closed model	69.95	69.94	69.75	66.62	69.68	70.76	65.52	70.80	76.57
5	Qwen3-VL-8B	Open model	67.36	68.93	64.70	63.38	65.59	71.40	61.46	67.34	76.08
6	MiMo-VL-7B-RL	Open model	60.55	57.63	56.60	51.93	65.90	71.34	52.83	57.64	70.55
7	Kimi-VL-A3B	Open model	58.58	62.98	52.85	50.72	54.94	73.91	55.96	52.67	64.64
8	GLM-4.1V-9B	Open model	58.56	55.19	56.63	55.52	54.68	63.83	54.66	59.31	68.70
9	MiniCPM-V4.5-8B	Open model	52.60	52.33	42.69	44.54	39.46	72.41	47.46	56.44	65.49
10	InternVL3.5-8B	Open model	49.38	54.26	37.83	40.97	39.43	79.46	43.64	49.41	50.06
11	GPT-5	Closed model	47.39	39.61	31.97	37.64	17.47	80.64	50.09	57.50	64.24
12	GPT-4o	Closed model	46.69	39.66	30.82	35.31	16.67	78.51	48.09	57.08	67.35
13	Ministral-3-8B	Open model	41.58	25.21	35.80	24.42	26.18	58.54	49.01	55.66	57.82
14	Claude-Sonnet-4.5	Closed model	39.71	37.59	31.20	29.12	20.88	41.64	49.22	44.18	63.87
15	Gemma-3-12B	Open model	30.90	25.71	14.17	17.36	9.42	76.24	38.32	30.23	35.77
16	SmolVLM2-2.2B	Open model	12.83	5.74	1.51	3.05	2.15	36.49	11.64	19.73	22.36

Score: macro F1 over 8 scenarios. Ret. receipts; Inv. invoices; Con. contracts; zh Chinese, en English. Underlined: best in the column. Equal scores share a rank.

UNIKIE-BENCH (ACL 2026), open-category table (15 published systems); every input capped at 1,605,632 pixels. DocQuorum: quorum over three readings from Qwen3.8-27B, then zoom re-reading. DocQuorum: the final pipeline.

ExtractBench: complete results (1/3)

#	System	Type	Score	Short	Medium	Long
1	LlamaExtract Agentic Plus	Commercial service	95.91	96.94	94.45	90.18
1	Pulse (Effort)	Commercial service	95.91	96.46	95.01	93.51
3	**DocQuorum (ours)**	**Open models**	**95.43**	**96.34**	**94.11**	**90.41**
4	LlamaExtract Agentic	Commercial service	94.77	96.04	92.11	91.83
5	Codex (GPT-6 Sol Evidence)	Coding agent	94.57	95.99	92.42	87.25
6	Codex (GPT-5.6 Sol Evidence)	Coding agent	93.77	96.04	90.19	82.73
7	Codex (GPT-5.5)	Coding agent	93.57	95.68	91.15	78.88
8	Codex (GPT-5.5 Evidence)	Coding agent	93.35	95.59	88.72	87.97
9	Pulse (Non-effort)	Commercial service	92.70	93.17	92.22	89.07
10	LlamaExtract Cost-Effective	Commercial service	92.27	94.03	89.51	83.73
11	Codex (GPT-5.6 Terra Evidence)	Coding agent	92.26	95.60	86.69	77.36
12	OpenAI GPT-6 Astra	Closed model	91.91	97.22	90.56	31.70
13	Codex (GPT-5.6 Luna Evidence)	Coding agent	91.16	93.50	87.43	80.05
14	Codex (GPT-6 Luna Evidence)	Coding agent	91.01	93.09	87.71	80.88
15	Reducto Deep Extract	Commercial service	90.44	94.20	80.47	92.01
16	Qwen3.8 Flash Next	Open model	89.88	94.82	87.81	37.74

Score: value F1, mean over all documents. Short, Medium, Long: the three splits (252, 98, 20 documents). Underlined: best in the column. Equal scores share a rank.

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems). ExtractBench served as the development set. DocQuorum: the final pipeline.

ExtractBench: complete results (2/3)

#	System	Type	Score	Short	Medium	Long
17	Qwen3.8 27B	Open model	89.75	94.68	87.54	38.45
18	Extend (Max Context)	Commercial service	88.62	92.32	78.75	90.46
19	Qwen3.6 35B	Open model	88.11	92.86	86.37	36.90
20	Qwen3.5 35B	Open model	88.00	93.52	85.27	31.73
21	DeepSeek V4.1 Flash	Closed model	87.11	94.44	81.49	22.23
22	Claude Code (Opus 4.8)	Coding agent	87.09	90.08	79.21	88.07
23	Claude Code (Opus 4.8 Evidence)	Coding agent	86.84	90.54	79.17	77.94
24	Qwen3.5 9B	Open model	86.10	92.31	81.12	32.22
25	Claude Code (Opus 5.5 Evidence)	Coding agent	86.06	89.63	77.96	80.83
26	Extend Extract	Commercial service	85.72	89.39	84.14	47.21
27	Datalab (Accurate + Balanced)	Commercial service	85.70	89.40	85.08	42.04
28	Reducto Extract	Commercial service	83.41	92.97	69.59	30.65
29	Kimi K3	Closed model	83.17	94.64	69.64	4.99
30	Qwen3.5 4B	Open model	82.43	89.06	76.64	27.31
31	NuExtract3	Open model	81.02	86.41	77.01	32.71
32	GLM-5.3 Flash	Closed model	80.75	96.30	51.56	27.83

Score: value F1, mean over all documents. Short, Medium, Long: the three splits (252, 98, 20 documents). Underlined: best in the column. Equal scores share a rank.

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems). ExtractBench served as the development set. DocQuorum: the final pipeline.

ExtractBench: complete results (3/3)

#	System	Type	Score	Short	Medium	Long
33	Google Gemini 3.8 Flash	Closed model	80.71	88.14	72.23	28.70
34	Google Gemini 3.5 Flash	Closed model	79.84	87.87	69.76	27.90
35	Lift Datalab 9B	Open model	78.43	87.64	65.69	24.78
36	OpenAI GPT-5.4 Nano	Closed model	74.90	77.43	76.37	35.81
37	Gemma4 26B	Open model	70.27	77.70	59.61	28.88
38	Gemma4 E4B	Open model	69.64	81.10	48.93	26.82
39	InternVL3.5 14B	Open model	66.35	80.70	39.04	19.41
40	InternVL3.5 8B	Open model	64.63	79.26	36.62	17.57
41	Qwen3.5 2B	Open model	63.40	71.22	50.93	25.93
42	Kimi-VL A3B	Open model	57.77	69.86	34.61	18.94
43	Gemma4 E2B	Open model	51.85	63.71	28.96	14.58
44	NuExtract 2.0 8B	Open model	49.42	62.92	23.50	6.39
45	Qwen3.5 0.8B	Open model	46.63	57.44	26.98	6.65
46	MiniCPM-V 4.5	Open model	45.85	57.39	25.30	1.13
47	Gemma4 12B	Open model	45.39	53.62	30.82	13.16
48	GLM-4.6V Flash	Open model	45.19	47.28	44.52	22.17

Score: value F1, mean over all documents. Short, Medium, Long: the three splits (252, 98, 20 documents). Underlined: best in the column. Equal scores share a rank.

ExtractBench leaderboard, snapshot of 2 October 2026 (47 published systems). ExtractBench served as the development set. DocQuorum: the final pipeline.

DocuBench: complete results

#	System	Type	Score
1	DocuPipe (high effort)	Commercial service	97.02
2	DocuPipe (standard effort)	Commercial service	96.14
3	Claude Sonnet 5 (direct LLM)	Closed model	91.73
4	**DocQuorum (ours)**	**Open models**	**91.54**
5	Reducto Deep Extract	Commercial service	89.38
6	Reducto (standard)	Commercial service	81.11
7	Extend	Commercial service	80.28
8	GPT-5.5 (direct LLM)	Closed model	76.48
9	Gemini 3.5 Flash (direct LLM)	Closed model	72.98
10	Unstructured	Commercial service	67.67

Score: macro field accuracy by the benchmark's own scorer, over PDFs, scans, photos, spreadsheets, Word, HTML, XML, CSV and text files. Equal scores share a rank.

DocuBench v0.2.0 leaderboard (9 systems). DocQuorum: the final pipeline.